Design Thinking and Social Environment for Leaf Disease Detection using ML

1. Empathize: Understand farmers' problems such as difficulty in identifying diseases, lack of expert support, and crop loss.

2. Define: Farmers need a simple and fast way to detect leaf diseases to reduce crop loss.

3. Ideate: Solutions include mobile apps, web systems, and ML-based detection using images.

4. Prototype: Build ML models using datasets of leaf images, preprocessing, and classification algorithms like CNN.

5. Test: Evaluate accuracy, precision, recall, and improve based on user feedback.

Social Environment Impact:

• Helps farmers detect diseases early and improve productivity.

• Increases income and reduces dependency on experts.

• Promotes eco-friendly farming by reducing pesticide use.

• Challenges include digital literacy and internet access.

Technologies Used: Python, Machine Learning, OpenCV, TensorFlow, Scikit-learn.

Conclusion: The system improves agriculture efficiency and supports sustainable farming.